

基礎生物学研究所

ゲノムインフォマティクス・トレーニングコース2026年3月

ゲノム情報の可視化

Visualize Genomes

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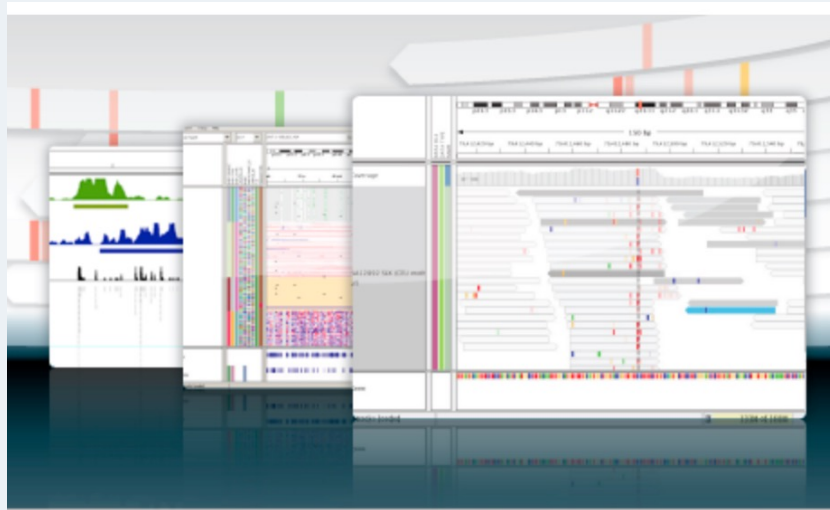
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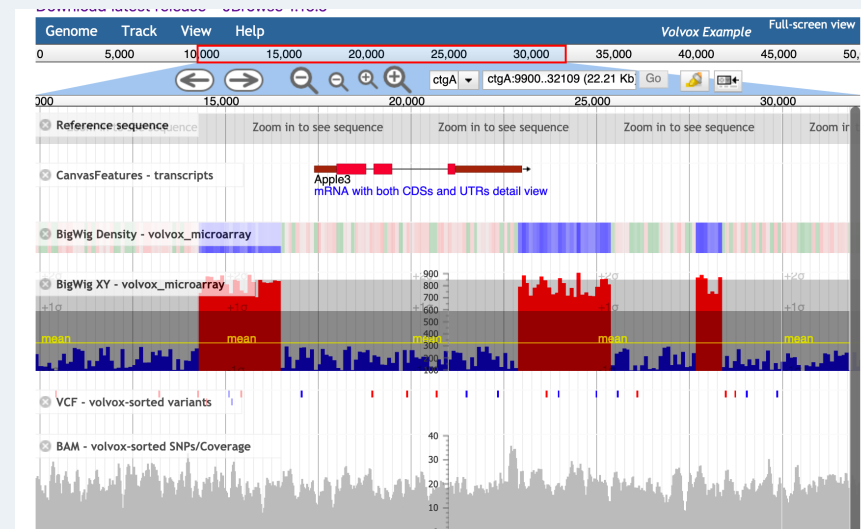
ゲノムブラウザ / Genome Browser

- ▶ ゲノムブラウザ (genome browser) は、ゲノム配列上の遺伝子やその他の機能要素を視覚的に表示し、検索・探索を可能にするツールやウェブサービス。
- ▶ 遺伝子構造に関する情報や、その遺伝子に関連するアノテーション情報 (機能や発現データなど) を閲覧。
- ▶ ヒトやモデル生物では多数の公開ゲノムブラウザが利用可能。自前で構築することも可能。

Local



Server



JBrowse2

JBrowse2 is a modern, extensible platform for visualizing and exploring genome-scale data.

- ▶ Interactive genome browser for exploring genomic data in a web browser
- ▶ Displays many types of genomic tracks (genes, alignments, variants, etc.)
- ▶ Supports common formats such as BAM, VCF, GFF, BED, BigWig
- ▶ Provides multiple visualization modes (linear view, circular view, synteny view)
- ▶ No complex server setup required — can be hosted as static files without a relational database
- ▶ JavaScript-based architecture — runs directly in modern web browsers
- ▶ Extensible via plugins for new tracks, views, and data sources

<https://jbrowse.org/jb2/>

SOFTWARE

Open Access



JBrowse: a dynamic web platform for genome visualization and analysis

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Abstract

Background: JBrowse is a fast and full-featured genome browser built with JavaScript and HTML5. It is easily embedded into websites or apps but can also be served as a standalone web page.

Results: Overall improvements to speed and scalability are accompanied by specific enhancements that support complex interactive queries on large track sets. Analysis functions can readily be added using the plugin framework; most visual aspects of tracks can also be customized, along with clicks, mouseovers, menus, and popup boxes. JBrowse can also be used to browse local annotation files offline and to generate high-resolution figures for publication.

Conclusions: JBrowse is a mature web application suitable for genome visualization and analysis.

Keywords: Genome, Browser, Bioinformatics

Single Views

A Linear Genome View



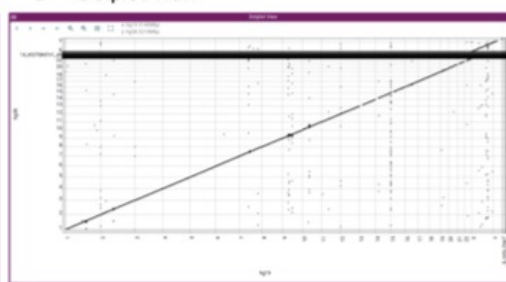
B Linear Genome View (Overview)



C Circular View



D Dotplot View



E Tabular View

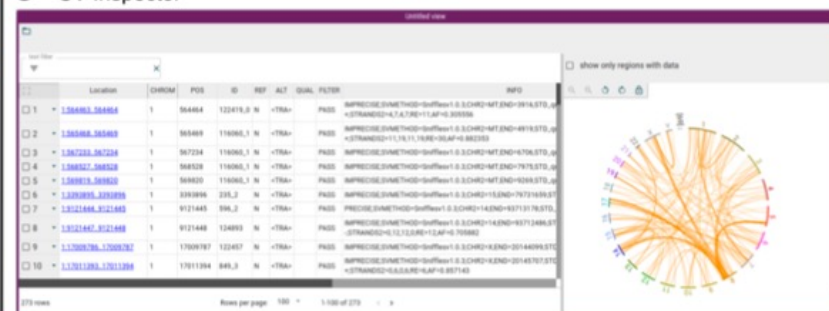


Combination Views

F Linear Synteny View



G SV Inspector



H Breakpoint Split View



Take-home message

- ▶ 2大ゲノムブラウザ: IGVとJBrowse
- ▶ ゲノムブラウザは、膨大なゲノム情報を視覚的に整理するとともに、多様なデータを統合的に参照するための強力な基盤である。